SYNTHETIC DEMO PAPER. Fictional content created only to test PaperPilot.

Continuous sEMG-Guided Grip Rehabilitation in Chronic Stroke: A Pilot Study

Fictional Author A, Fictional Author B (Synthetic Demo Institute), 2026

Abstract

We show that a grip rehabilitation protocol guided by continuous sEMG feedback improved hand function in stroke patients. All data were recorded in Dataset A.

1. Introduction

Hand weakness is common after stroke. Many rehabilitation protocols rely on strength scores only.

2. Methodology

Participants were stroke patients treated in a single clinic. Continuous sEMG was recorded during every grip training session and stored in Dataset A.

3. Results

Hand function scores improved significantly in stroke patients after the grip rehabilitation protocol. Muscle activation increased significantly during grip tasks measured with continuous sEMG. Grip strength also improved in stroke patients.

4. Discussion and Limitations

Limitations include a small sample and a single clinic. The protocol was not tested in other patient groups.

5. Conclusion

Continuous sEMG feedback supports recovery when the grip rehabilitation protocol is used.

References

[1] Fictional A. et al. Synthetic reference. 2024.